

Adaptive Hybrid Learning for Optimal Control¹

Master's Thesis

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¹novel term, inspired from [1]

Abstract and Outline

1. Problem and Research Question
2. Methods and Algorithm
3. Experiments
4. Conclusion
5. Discussion

Optimal Control Problem

Find $\mathbf{u}(t) \in \mathcal{U} := \{\mathbf{u} : [0, \infty) \rightarrow U \text{ measurable}\}$ **stabilizing** the system

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t)) + B\mathbf{u}(t), \quad \mathbf{x}(0) = x_0 \in \mathbb{R}^n, \quad t \in [0, \infty),$$

while minimizing

$$J(x_0; \mathbf{u}) := \int_0^\infty (\mathbf{x}^\top Q \mathbf{x} + \mathbf{u}^\top R \mathbf{u}) dt, \quad Q \succ 0, R \succ 0.$$

Ch. 2 (Theory): $U \subset \mathbb{R}^m$ compact

→ enables self-contained derivation of existence, uniqueness, and stability as in [4]

Ch. 3 (Methods): $U = \mathbb{R}^m$

→ quadratic structure admits closed-form minimizer $u^*(x)$

→ enables the LQR baseline and GradQRnet parameterization

Assumptions

(A1) Local Lipschitz: $\forall K \subseteq \mathbb{R}^n$ compact, $\exists L_K > 0 : \|f(x) - f(y)\| \leq L_K \|x - y\|, \forall x, y \in K$

(A2) Dissipativity: $(f(x) - f(y))^\top (x - y) \leq L \|x - y\|^2, \forall x, y \in \mathbb{R}^n$

(A3) Stable origin: $f(0) = 0$

(A4) Existence of a stabilizing control $\mathbf{u} : J(x_0; \mathbf{u}) < \infty$

(A5) Additional regularity: $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$

- A1–A3: Regularity conditions ensuring well-posedness of the dynamics
- A2 can be weakened to local (semiglobal) condition, but again kept for a complete, self-contained illustration, and clean, global treatment
- A4 redundant given A2, but stated independently since A2 fails for Allen-Cahn and KS, and it simplifies several proofs
- A5 is needed for Pontryagin's Principle and the gradient-costate-equality

Solution

A1, A3 and A4: Optimal control u^* and value function $V(x)$ exist

+ A2: V is the unique viscosity solution ² to the HJB-equation:

$$\min_{u \in U} H(x, u, \nabla V) = 0, \quad H(x, u, \nabla V) := x^\top Qx + u^\top Ru + \nabla V(x)^\top (f(x) + Bu)$$

+ $U = \mathbb{R}^m$: u^* admits closed-loop form $u(x) = -\frac{1}{2}R^{-1}B^\top \nabla V(x)$

+ A5: A1 becomes redundant and Gradient-Costate-Identity holds

Bottleneck: Curse of Dimensionality

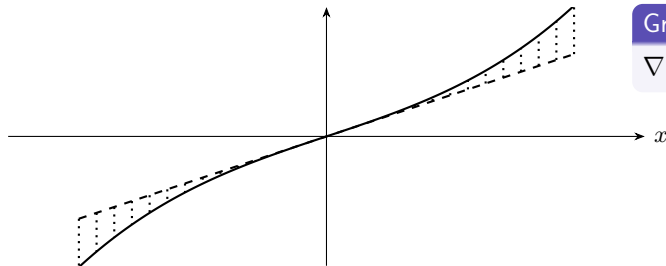
Classical numerical methods require N grid points per axis $\implies$ complexity $\mathcal{O}(N^n)$

Solution: Neural Network Approximation

$$\nabla V(x) \approx \nabla V_\theta(x) \implies u_\theta(x) = -\frac{1}{2}R^{-1}B^\top \nabla V_\theta(x)$$

²Section 2.3.2 contains a definition, thorough explanation and worked example

LQR-Residual Model



GradQRnet [7]

$$\nabla V_{\theta}(x) = \nabla V_{\text{LQR}}(x) + \mathcal{N}_{\theta}(x) - \mathcal{N}_{\theta}(0)$$

--- ∇V_{LQR} — ∇V_{θ} $\mathcal{N}_{\theta}$

- $\mathcal{N}_{\theta}$ is a feedforward network with 3 hidden layers, width $4n$ and $\tanh$ activation³
- Built-in stability: $\mathcal{N}_{\theta}(x) - \mathcal{N}_{\theta}(0)$ ensures $\nabla V_{\theta}(0) = 0$ and $u_{\theta} \approx u_{\text{LQR}}$ near the target
- Better conditioning: small corrections to LQR are easier learnable than full gradient ∇V

³The thesis mistakenly mentions softplus instead, which lacks boundedness needed for later arguments

Data Generation [5]

A2–A5: For given x_0 , optimal trajectories satisfy Pontryagin's Principle

$$\begin{cases} \mathbf{u}^*(t) = \arg \min_{\mathbf{u}} H(\mathbf{x}^*, \mathbf{u}, \boldsymbol{\lambda}^*), \\ \dot{\mathbf{x}}^*(t) = f(\mathbf{x}^*) + B\mathbf{u}^*, \quad \mathbf{x}^*(0) = x_0, \\ -\dot{\boldsymbol{\lambda}}^*(t) = 2Q\mathbf{x}^* + \nabla f(\mathbf{x}^*)^\top \boldsymbol{\lambda}^*, \quad \lim_{t \rightarrow \infty} \boldsymbol{\lambda}^*(t) = 0. \end{cases}$$

and the Gradient-Costate-Identity

$$\boldsymbol{\lambda}^*(t) = \nabla V(\mathbf{x}^*(t)) \quad \text{for a.e. } t \geq 0.$$

Strategy [7]

1. Sample N_{traj} initial conditions $\{x_0^{(k)}\}$
2. Solve PMP-BVP on each $x_0^{(k)}$ to obtain open-loop trajectories
3. Collect state-costate pairs into $\mathcal{D} = \{(\mathbf{x}^{(i)}, \boldsymbol{\lambda}^{(i)})\}_{i=1}^{|\mathcal{D}|}$ and learn θ on $\mathcal{D}$

Option 1: Supervised Loss [7]

$$\mathcal{L}_{\text{sup}}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(x, \lambda) \in \mathcal{D}} \|\nabla V_{\theta}(x) - \lambda\|^2 \approx \mathbb{E}_{x_0 \sim \Omega} [\|\nabla V_{\theta}(x) - \lambda\|^2] \quad (\text{offline, fixed data})$$

- + Strong gradient signal
- Costly data generation and limited coverage/distribution mismatch

Option 2: Unsupervised Loss [2], [3]

$$\mathcal{L}_{\text{HJB}}(\theta) = \frac{1}{|\mathcal{B}|} \sum_{x_0 \in \mathcal{B}} \sum_{t=0}^{h-1} \mathcal{H}(x_t, \nabla V_{\theta}(x_t))^2 \Delta t_t \approx \mathbb{E}_{x \sim \rho_{\theta}} [\mathcal{H}(x, \nabla V_{\theta}(x))^2] \quad (\text{online, policy-induced})$$

- + No expert data needed
- Non-convex, prone to local minima

Option 3: Combined (“Hybrid”) Loss

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{HJB}}(\theta) + \lambda_{\text{sup}} \mathcal{L}_{\text{sup}}(\theta)$$

Curriculum Annealing: $\lambda_{\text{sup}}(e) = \lambda_{\text{sup}}^{(0)} \cdot \max(0, 1 - \frac{e}{E})$

- Supervised term anchors network to ground-truth costate data; annealing prevents catastrophic forgetting during unsupervised phase
- Outperforms both pure pretraining (hard two-level switch) and fixed λ_{sup}

Adaptive time stepping: $\Delta t(x) \propto \frac{1}{\|f(x)\|} \in [\Delta t_{\min}, \Delta t_{\max}]$

- Smaller steps far from origin, larger steps near equilibrium where HJB residual is small
- Mentioned in [3]; this work proposes explicit rule matching BVP solver scaling

Adaptive LQR-Residual Sampling

Both training modes require sampling initial conditions x_0 :

Uniform: $x_0 \sim \mathcal{U}(\Omega)$

Adaptive [6]: $\|\nabla V_\theta(x_0)\|$ large

Research Gap [3]: Can adaptive sampling benefit *unsupervised* and *hybrid* learning?

Empirically: sampling via $\|\nabla V_\theta(x_0)\|$ performed poorly $\rightarrow$ states too hard for untrained u_θ

Theoretically: $\|\nabla V_\theta(x)\| = \|2Px + \mathcal{N}_\theta(x) - \mathcal{N}_\theta(0)\| \lesssim \|2Px\| + \mathcal{O}(1)$

- $\rightarrow \|2Px\|$ dominates for large $\|x\|$ for any bounded network
- $\rightarrow$ selects far states regardless of approximation quality (hard for untrained network)

Novel idea: Adaptive LQR-Residual Sampling

sample and select x_0 where $\|\mathcal{N}_\theta(x_0) - \mathcal{N}_\theta(0)\|$ large

- $\rightarrow$ tighter proxy for HJB residual than full gradient norm
- $\rightarrow$ targets states where learned correction is large

Adaptive Hybrid Optimal Control: Implementation

```
1 Sample  $N_{\text{traj}}$  initial conditions  $\{x_0^{(k)}\}$ 
2 Solve PMP-BVPs for each  $x_0^{(k)}$  and aggregate  $\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}}$ 

3 if supervised pretraining then
4   foreach  $\mathcal{B} \subseteq \mathcal{D}_{\text{train}}$  do
5      $\theta \leftarrow \text{Adam}(\theta, \nabla_{\theta} \mathcal{L}_{\text{sup}}(\mathcal{B}), \mu_{\text{sup}})$ 

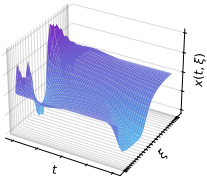
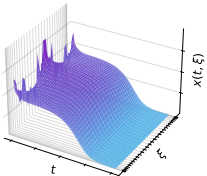
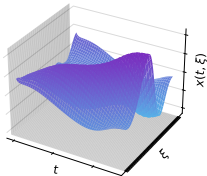
6 for  $s = 1, \dots, S$  do
7   if adaptive sampling then
8     Sample  $N_{\text{cand}} \times |\mathcal{B}|$  initial conditions  $\{x_0^{(j)}\}$ 
9     Select  $|\mathcal{B}|$  points  $x_0$  with highest residual norm  $\|\mathcal{N}_{\theta}(x_0) - \mathcal{N}_{\theta}(0)\|$ 
10  else
11    Sample  $|\mathcal{B}|$  initial conditions  $\{x_0^{(j)}\}$  uniformly
12    Rollout trajectories and set  $\mathcal{L} \leftarrow \mathcal{L}_{\text{HJB}}$ 
13    if supervised penalty then
14      Draw  $\mathcal{B} \subseteq \mathcal{D}_{\text{train}}$  and set  $\mathcal{L} \leftarrow \mathcal{L} + \lambda_{\text{sup}}(e) \mathcal{L}_{\text{sup}}(\mathcal{B})$ 
15     $\theta \leftarrow \text{Adam}(\theta, \nabla_{\theta} \mathcal{L}, \mu_{\text{unsup}})$ 
```

Experiment Setup

General form: Stabilize parabolic PDEs with localized bump actuators:

$$X_t = \mathcal{L}(X) + b(\xi)^\top u, \quad \min_u \int_0^\infty (\|X\|_{L^2}^2 + \tfrac{1}{2}|u|^2) dt$$

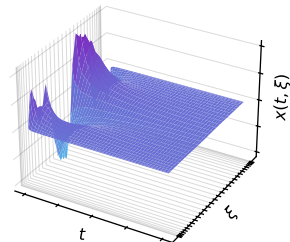
Discretized form: OCP (slide 3)

	Burgers	Allen–Cahn	Kuramoto–Sivashinsky
$\mathcal{L}(X)$	$\nu X_{\xi\xi} - \frac{1}{2}(X^2)_\xi + \alpha X e^{-\gamma X}$	$\nu X_{\xi\xi} + X(1 - X^2)$	$-\nu X_{\xi\xi\xi\xi} - X_{\xi\xi} - \frac{1}{2}(X^2)_\xi$
BC	Dirichlet	Neumann	periodic
			

Burgers Equation: Stability and Optimality

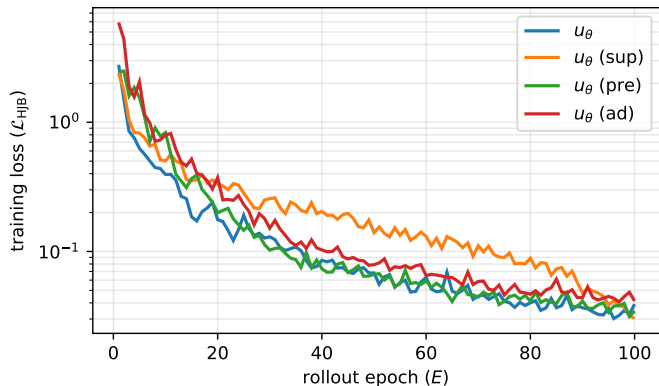
	u_{LQR}	u_θ	u_θ (sup)	u_θ (pre)	u_θ (ad)
Stability	$74 \pm 1\%$	$79 \pm 1\%$	$79 \pm 1\%$	$78 \pm 1\%$	$82 \pm 1\%$
Cost J	0.60 ± 0.01	0.63 ± 0.02	0.64 ± 0.02	0.61 ± 0.01	0.67 ± 0.02

	u_θ (pre/sup)	u_θ (sup/ad)	u_θ (pre/ad)	u_θ (pre/sup/ad)
Stability	$78 \pm 1\%$	$82 \pm 1\%$	$77 \pm 1\%$	$77 \pm 1\%$
Cost J	0.60 ± 0.01	0.67 ± 0.02	0.60 ± 0.01	0.61 ± 0.01



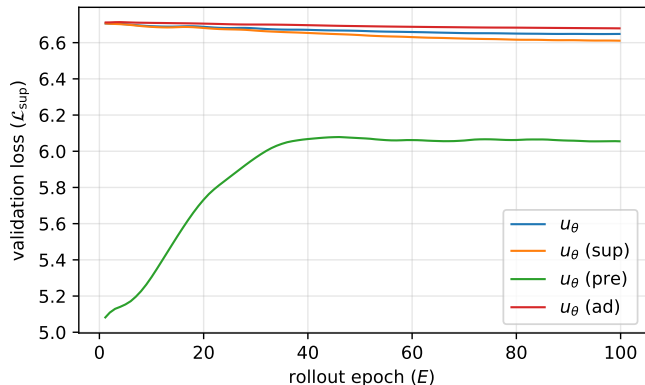
- Adaptive sampling (with and without supervised penalty) achieves highest stability
- Pretraining and adaptive sampling exhibit negative interaction

Training Dynamics



- $\mathcal{L}_{HJB}$ starts higher for modified models but eventually all converge with baseline
- Higher losses can be due to differences in sampling (ad) or learning objectives (sup, pre)

Training Dynamics



- $\mathcal{L}_{\text{sup}}$ roughly constant and much lower for pretrained controller $\rightarrow$ distribution mismatch
- Supervised controller keeps decreasing, in spite of $\lambda_{\text{sup}} \rightarrow 0$
- Validation error does not explain observed performance differences

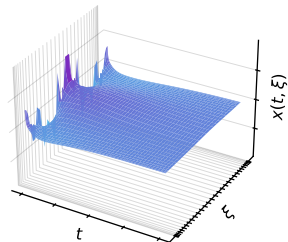
Allen-Cahn Equation: Generalization and Robustness

Viscosity mismatch: ν smaller

	u_{LQR}	u_θ	u_θ (sup)	u_θ (ad)	u_θ (sup/ad)
Stability	$45 \pm 2\%$	$49 \pm 2\%$	$50 \pm 2\%$	$52 \pm 2\%$	$54 \pm 2\%$
Cost J	0.81 ± 0.05	0.87 ± 0.05	0.87 ± 0.05	0.91 ± 0.05	0.99 ± 0.05

Stochastic perturbation: $+\sigma_{\text{noise}} \sim \mathcal{N}(0, I_n)$

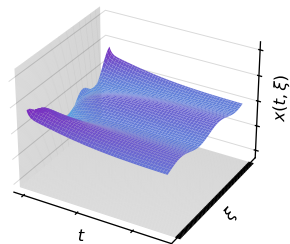
	u_{LQR}	u_θ	u_θ (sup)	u_θ (ad)	u_θ (sup/ad)
Stability	$74 \pm 1\%$	$71 \pm 1\%$	$70 \pm 1\%$	$71 \pm 1\%$	$73 \pm 1\%$
Cost J	2.00 ± 0.11	1.52 ± 0.08	1.52 ± 0.09	1.57 ± 0.09	1.65 ± 0.10



- Supervised penalty anchors learning near known optimal trajectories
- Adaptive sampling slowly extends coverage to nonlinear regime
- Hybrid controller (sup/ad) achieves highest stability both within and outside domain

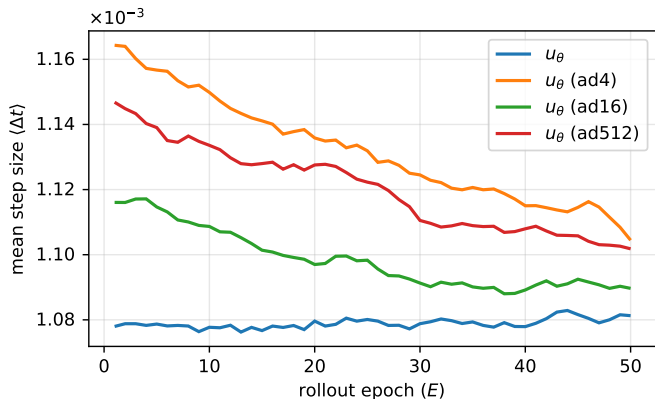
Kuramoto-Sivashinsky Equation: Data-Free Regime

	u_{LQR}	u_{θ}	u_{θ} (ad4)	u_{θ} (ad16)	u_{θ} (ad512)
Stability	$73 \pm 3\%$	$78 \pm 3\%$	$85 \pm 2\%$	$89 \pm 2\%$	$87 \pm 2\%$
Cost J	49.35 ± 1.82	49.60 ± 1.75	50.23 ± 1.69	49.54 ± 1.64	50.64 ± 1.67



- All adaptive variants substantially outperform LQR and baseline
- $N_{\text{cand}} = 16$: highest stability and lowest cost
- Non-monotone relationship: moderate selection pressure outperforms both weak ($N_{\text{cand}} = 4$) and aggressive ($N_{\text{cand}} = 512$)

Adaptive Step Size and Curriculum Effect



- $N_{\text{cand}} = 16$: moderate $\Delta t \rightarrow$ highest stability
- Moderate selection pressure places training near the informative frontier
- Supports hypothesis that adaptive time stepping and sampling interact favourably [3]

Conclusion and Outlook

Key findings:

1. Adaptive LQR-residual sampling consistently improves optimality and stability for unsupervised learning
2. Adding a supervised penalty further improves both metrics
3. Pretraining and adaptive sampling exhibit a negative interaction, potentially due to a distribution mismatch

Outlook:

- Convergence guarantees for the full learning pipeline
- Dynamic adjustment of N_{cand} via variance-to-gradient heuristics
- Adaptive scheduling of λ_{sup} based on training diagnostics

- Standing assumptions: scope and verification
- Activation function: $\tanh$ vs softplus
- Annealing vs two-level training ansatz
- Expressivity vs ansatz failure

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